



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Linux Networking Commands & Troubleshooting

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1.THEORY

1.1 Answer all 20 questions in your own words

1.What is a network interface?

A network interface is a connection that allows a computer to communicate with a network. It can be a physical adapter or a virtual interface.

2. What is the loopback interface?

The loopback interface is a special interface used by a computer to communicate with itself. It is commonly represented as lo and uses the address 127.0.0.1

3. What does ip addr display?

Ip addr displays network interfaces along with their IP addresses and other network configuration information.

4. What is the purpose of ip link?

Ip link shows information about network interfaces, such as whether an interface is up or down.

5. What does hostname -I display?

Hostname -I displays the IP addresses assigned to the computer.

6. What does ip route show?

Ip route shows the routing information of the system, including the default route used to reach other networks.

7. What is a default route?

A default route is the route used when the computer does not have a specific route for the destination network. It normally sends traffic through the default gateway.

8. What does ping test?

Ping tests whether a destination can be reached over a network and check basic connectivity.

9. Why can a failed ping be misleading?

A failed ping does not always mean the network or Internet is down. The destination or a firewall may block ICMP ping packets.

10. What is DNS?

Domain Name System converts domain names, such as google.com, into IP addresses that computer can use to communicate.

11. What does getent hosts do?

Getent hosts look up a hostname and displays the IP address associated with it using the system's configured name-resolution services.

12. What information does ip neigh display?

Ip neigh displays information about nearby network devices that the system has discovered, including their IP and MAC addresses.

13. What is a socket?

A socket is an endpoint used by applications to communicate over a network. It is associated with an IP address and a port.

14. What does `ss -tuln` display?

`ss -tuln` displays listening TCP and UDP network sockets, including the ports that are waiting for connections. The `-n` option shows addresses and ports as numbers.

15. What is the difference between TCP and UDP conceptually?

TCP is connection-oriented and provides reliable, ordered delivery of data. UDP is connectionless and faster but does not guarantee delivery or order.

16. What is curl used for?

Curl is a command-line tool used to communicate with web servers and other network services. It can be used to test or retrieve information from URLs.

17. What does `curl -I` request?

`Curl -I` requests only the HTTP response headers from a web server instead of downloading the full webpage.

18. What is traceroute used for?

Traceroute is used to show the network path taken by packets from your computer to a destination. It helps identify the hops between the source and destination.

19. Why is ss commonly preferred over netstat on modern Linux?

Ss is commonly preferred because it is a modern tool that is generally faster and provides detailed information about network sockets. Netstat is older and may not be installed by default.

20. Why are Linux networking commands useful in cybersecurity?

Linux networking commands help security professionals understand network interfaces, IP addresses, routes, connections, open ports, and network activity. They are useful for troubleshooting, monitoring and authorized security testing.

2.PRACTICAL SCREENSHOTS

2.1 (from your own VM/lab only): ip addr · ip route · ping -c 4 27.0.0.1 ·
getent hosts google.com · ss -tuln · curl -I https://example.com

```
(kali㉿kali)-[/home/kali]
PS> ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ad:25:87 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 557sec preferred_lft 557sec
    inet6 fe80::8619:d318:503e:8617/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali㉿kali)-[/home/kali]
PS> ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:ad:25:87 brd ff:ff:ff:ff:ff:ff

(kali㉿kali)-[/home/kali]
PS> ip route
>>
>> ^C

(kali㉿kali)-[/home/kali]
PS> ip route
192.168.56.0/24 dev eth0 proto kernel scope link src 192.168.56.102 metric 100

(kali㉿kali)-[/home/kali]
PS> ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.039 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.042 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.052 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.040 ms

— 127.0.0.1 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3096ms
rtt min/avg/max/mdev = 0.039/0.043/0.052/0.005 ms

(kali㉿kali)-[/home/kali]
PS> ping -c 4 google.com
/usr/bin/ping: google.com: Temporary failure in name resolution
```

```
(kali㉿kali)-[/home/kali]
PS> ping -c 4 google.com
/usr/bin/ping: google.com: Temporary failure in name resolution

(kali㉿kali)-[/home/kali]
PS> getent hosts google.com

(kali㉿kali)-[/home/kali]
PS> ip neigh
```

```
(kali㉿kali)-[/home/kali]
PS> ss -tuln
Netid      State      Recv-Q     Send-Q      Local Address:Port    Peer Address:Port

(kali㉿kali)-[/home/kali]
PS> cur; -I https://example.com
cur: The term 'cur' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
-I: The term '-I' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

(kali㉿kali)-[/home/kali]
PS> curl; -I https://example.com
curl: try 'curl --help' or 'curl --manual' for more information
-I: The term '-I' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

(kali㉿kali)-[/home/kali]
PS> curl -I https://example.com
curl: (6) Could not resolve host: example.com
```

2.2 TROUBLESHOOTING QUESTION (from lab Task 10):

A Linux system can ping 8.8.8.8, but ping google.com fails because the hostname cannot be resolved. What type of problem is most likely, and which command from today would you use first to confirm it?

A Linux system can ping 8.8.8.8, but ping google.com fails because the hostname cannot be resolved. The problem might be DNS problem/ DNS resolution problem.

8.8.8.8 is an IP address, so the system can reach the Internet using an IP.

But google.com is a hostname. The system needs DNS to convert.

The first command I would use is: `getent hosts google.com`